Week 2 Control a character · paper route

Paper route · GROW Computing · 40 minutes

Name: ____________________ Date: __________

Week 2. Control a character.

This is your Week 2 work. It is not a practice.

You do not need a computer for this.

You need this pack, a pen and one counter.

A coin or a rubber works as a counter.

You will write two control scripts.

You will test them by moving your counter.

Moving your counter one square is a key press.

That is how you run a program on paper.

Your teacher watches you do it.

Your teacher writes down what you did.

That is the same record as the computer route.

Point, say, draw or write. All four count.

An adult can write your exact words for you.

At the end you name this pack and file it.

You find it again and check your work is still right.

Nothing is left to do later on a computer.

Point, say, draw or write. Someone may record your exact words.

On the screen route this week is saved as W02_MyControls.sb3. On the paper route no file is made. Your pages are the record: write your name and the date on every page and put them in your folder. An adult initials the folder. Nobody should ask you for W02_MyControls.sb3 if you took the paper route.

Write your code

Courier · The reset script. This one is already made for you. It is the paper copy of W02_Start.sb3. Copy the two blocks into the rows so you know what the green flag does. Copying this script is not one of your own controls.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- · go to x: 0 y: 0
- · point in direction 90
- · change x by 10
- · change x by -10
- · change y by 10
- · change y by -10

Courier · Your right control. This is the first script you build. Ring the word right in the hat to show you chose that key. Then write the one block that must run under it. Choose it from the word bank.

Write one block on each line. Start with the hat block, which is done for you.

when right arrow key pressed

Blocks you may use this week:

- · go to x: 0 y: 0
- · point in direction 90
- · change x by 10
- · change x by -10
- · change y by 10
- · change y by -10
- · change x by ___   (you write the number)

Courier · Your left control. This is the second script you build. It is also the repair for the faulty script in Trace 1. Ring the word left in the hat. Watch the minus sign.

Write one block on each line. Start with the hat block, which is done for you.

when left arrow key pressed

Blocks you may use this week:

- go to x: 0 y: 0
- point in direction 90
- change x by 10
- change x by -10
- change y by 10
- change y by -10

Courier · The up control. Your teacher shows this one. Build it if you are on the Challenge route. Up and down change y, not x.

Write one block on each line. Start with the hat block, which is done for you.

when up arrow key pressed

Blocks you may use this week:

- go to x: 0 y: 0
- point in direction 90
- change x by 10
- change x by -10
- change y by 10
- change y by -10

Courier · The down control. Your teacher shows this one. Build it if you are on the Challenge route. Down uses a minus sign, like left.

Write one block on each line. Start with the hat block, which is done for you.

when down arrow key pressed

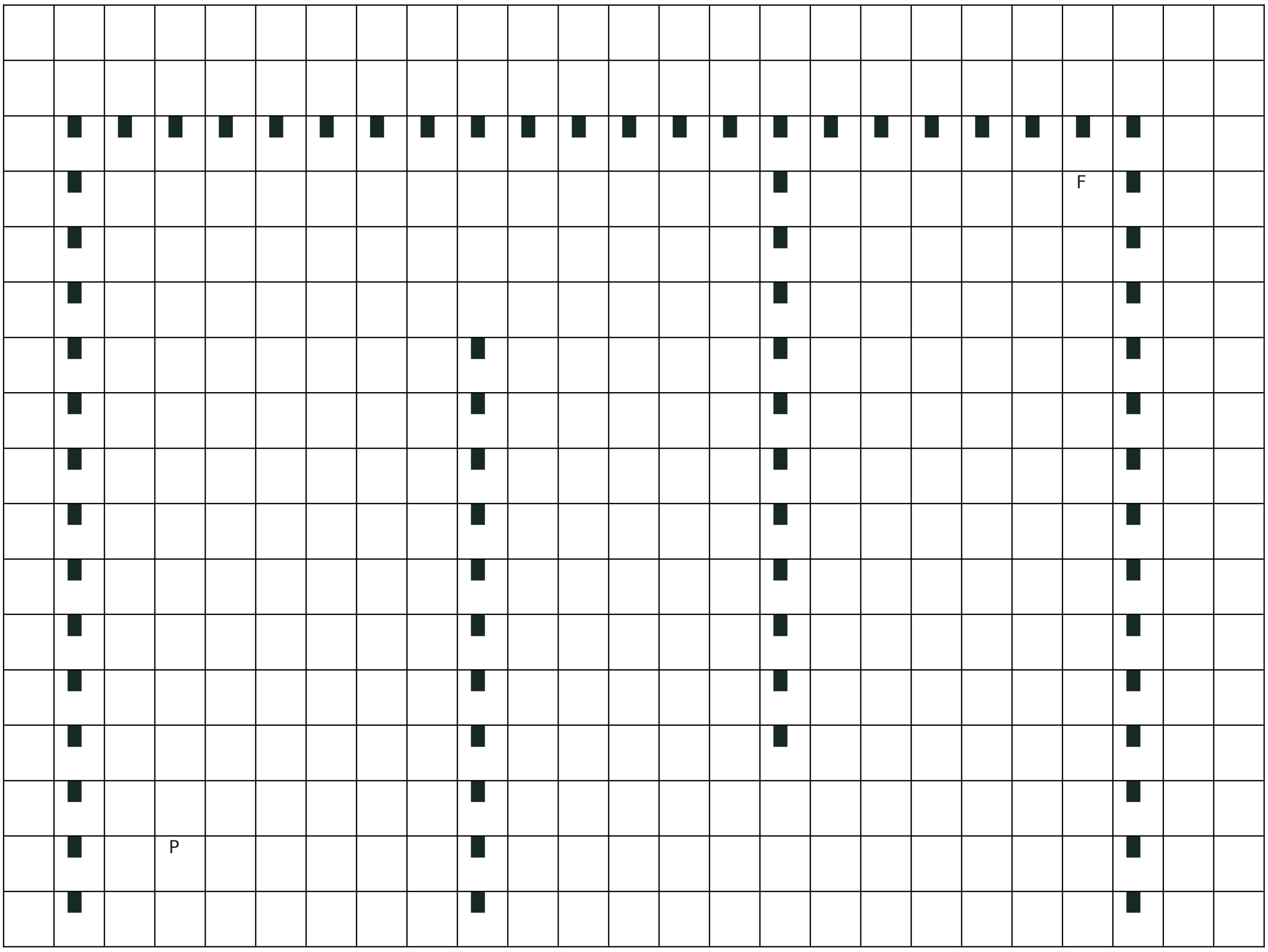
Blocks you may use this week:

- go to x: 0 y: 0
- point in direction 90
- change x by 10
- change x by -10
- change y by 10
- change y by -10

The maze

The printed maze is the stage. It is 480 wide and 360 tall. x goes from -240 to 240. y goes from -180 to 180. The squares are 10 apart. One key press moves your counter one square. The walls are printed solid. A counter cannot sit on a wall. Two squares are marked. The GREEN FLAG SQUARE is x = 0, y = 0, in the middle corridor. That is where the green flag puts Courier this week. The MAZE HOME square is x = -180, y = -120, at the bottom left. That is where the player starts in the real maze. You use the maze four times this week. One. Put your counter on the green flag square. Say right, right, left, left. Move one square each time. Fill in Trace 2. Two. Put your counter back on the MAZE HOME square. Use your right control only. Move one square right at a time along the corridor. Stop on the last free square before the wall. Mark that square with a cross. Fill in Trace 3. Three. Now use your left control. Count the presses back to the MAZE HOME square. Four, on the Challenge route. Put your counter on the green flag square. Move to the beacon at x = 20, y = 20 in four presses. Then do it again in a different order. The finish flag at x = 180, y = 120 is not for this week. You cannot reach it with right and left only. The wall stops you. You will need up and down. Those are checked in the maze in Week 4.

The maze below is the real one, at its real size. One square is 20 steps. Player starts at (-180, -120). Finish is at (180, 120). Move a counter square by square and write down where it lands.



Predict, trace, compare

Trace 1. The faulty left script. This script is printed on your fault card. The hat says when left arrow key pressed. The block under it says change x by 10. Put your counter on the green flag square, x = 0, y = 0. Move your counter for every row. Write what x and y become.

Step	Block that runs	x after	y after
Green flag	go to x: 0 y: 0	0	0
Green flag	point in direction 90	0	0
Press left once	change x by 10		
Press left once more	change x by 10		
Which block is wrong? Ring it above.	It should say:		

Predict then check. Your repaired left script. First write your repair into the left arrow grid. Then put your counter back on the green flag square, x = 0, y = 0. Write what you expect first. Then move your counter. Then write what you got.

Press once	I expect x to be	x after I move my counter	Same? tick or cross
Left			
Then right			

Trace 2. Right, right, left, left. Put your counter on the green flag square, x = 0, y = 0. One press is one square. Press and let go. Do not slide the counter two squares.

Step	Block that runs	x after	y after
Green flag	go to x: 0 y: 0	0	0
Press right (1)	change x by 10		
Press right (2)	change x by 10		
Press left (1)	change x by -10		
Press left (2)	change x by -10		

Trace 3. The maze corridor. Put your counter on the MAZE HOME square, x = -180, y = -120. Use your right control only. Rows 1 and 2 are done for you. Keep going until the wall stops you.

Move	x after	y after	Free square or wall?
Start on the MAZE HOME square	-180	-120	free
Right press 1	-170	-120	free
Right press 2			
Right press 3			
Right press 10			
Right press 11 would land on			
Left presses back to MAZE HOME	-180	-120	how many?

Challenge A. Two orders to the beacon. Put your counter on the green flag square, x = 0, y = 0. The beacon is x = 20, y = 20. Use four presses each time. Go back to the green flag square before you try the next order.

My four presses, in order	x at the end	y at the end	Reached the beacon? tick or cross
right, right, up, up			
up, right, up, right			
My own order:			

Challenge B. Smaller moves. Do this one in the table only. Leave your right and left grids at 10 and -10. For this table only, right changes x by 5 and left changes x by -5. Predict first: how many right presses reach x = 20? Write your prediction here: ______

Press	x after	y after
Green flag	0	0
Right press 1		
Right press 2		
Right press 3		

Right press 4		
Green flag again	0	0
Right press		
Right press		
Left press		
Left press		

My completion record

What this week shows

Week 2 assessment destination, in the pack's own words: Foundation practice. "These lessons build foundations for AQA 71638 Programming with Scratch (unit 6), Level One. Neither Week 1 nor Week 2 completes the full unit." No AQA outcome is claimed this week. The work here evidences the pack's own core completion statement for Week 2: "The pupil creates and tests BOTH right and left scripts. One working script is an early milestone and leaves a recorded next step. All four controls are checked in the main maze in Week 4." The shared goal is: "I can use key events to move a sprite and fix a direction mistake." This feeds outcome 2, "Create keyboard arrow-control scripts", which is claimed in Week 4 and not before.

☐ I can point, say, draw or write every answer below. An adult can write my words for me.

☐ I wrote a right arrow script. It says change x by 10.

☐ I wrote a left arrow script. It says change x by -10.

☐ I found the wrong block in the faulty script. I showed what it should say instead.

☐ I put my counter on the green flag square. I traced right, right, left, left. I finished on 0.

☐ I put my counter on the MAZE HOME square. I moved it right along the corridor. I stopped on the last free square before the wall.

☐ I can show what the minus sign does.

☐ I wrote my name and W02_MyControls on the front of this pack.

☐ I put my pack in my folder.

☐ I found my pack again. I read my two scripts. I traced one of them again. I got the same finish.

☐ I said one idea for my mission.

Adult record

☐ Create a right-key script: observed the pupil write change x by 10 into the right arrow grid, chosen from a word bank that also offered change x by -10, change y by 10 and change y by -10. Record the pupil choosing, not the pupil copying.

☐ Create a left-key script using -10: observed the pupil write change x by -10 into the left arrow grid.

☐ Explain or diagnose the wrong-sign bug: observed the pupil ring change x by 10 under the left arrow hat on the fault card, and give the repair by saying it, by pointing to the correct repair card, or by writing -10.

☐ Reset and test right right left left: observed the pupil return the counter to the green flag square and move it one square per press, recording x = 10, 20, 10, 0 in Trace 2. Tracing the counter is the practical demonstration on this route.

□ Save, locate and reopen the controls: observed the pupil name and date the pack as W02_MyControls, file it, find it again unaided, reread both grids and retrace one control to the same finish. Write down what actually happened. Do not write "saved and reopened an .sb3" for a pupil on the paper route.

□ Support record: independent / visual cue / general prompt / precise model / physical access help / inherited code. On this route, "inherited code" means the printed ready-made controls card was handed over. A ready-made card does not establish pupil authorship, exactly as a recovery .sb3 does not.

□ Route record: write in this teacher guide that the pupil completed Week 2 by the paper route. Both routes evidence the same core completion. Do not write on, change or reprint the AQA summary sheet - Week 2 claims no outcome, so there is nothing to enter on it.

□ Readiness review: keep any missing second script as a named next step. All four pupil-created controls remain due in the main maze in Week 4.

Adult name: ____________________ Date: __________

Route taken: paper · Recorded in the teacher guide, not on the AQA form.